

KANAV ATRE

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Software engineer with internship and research experience building large-scale data and ML-backed systems. Experienced across the stack with strong focus on reliability, performance, and automation.

EDUCATION

Purdue University

Bachelor of Science Double Major in Data Science & Applied Statistics

West Lafayette, IN

Expected Dec 2026

- **Coursework:** Data Mining & Machine Learning, Intro to AI, Analysis of Algorithms, Relational Databases
- **Involvement:** 2 x presenter at **Purdue's Undergraduate Research Conference**; Purdue LaunchPad **Project Lead**; Student Ambassador for Purdue VIP

EXPERIENCE

Software Engineer Intern

May 2025 – Aug 2025

KetteQ

Atlanta, GA

- Built a real-time migration pipeline for **10M+ CRM records** from Neon to ClickHouse
- Developed automated validation and testing scripts in Python, reducing manual QA effort by **10+ hrs/week**.
- Wrote & Optimized large-scale **PostgreSQL** queries, enabling **sub-second** refresh
- Deployed CDC snapshotting, archival scripts, and visualizations using **Kafka & Apache ECharts**
- Attended daily scrums; worked and delivered in a **fast paced Agile** engineering team

Undergraduate ML Researcher

Aug 2023 – May 2024

Purdue VIP (Vertically Integrated Projects)

West Lafayette, IN

- Built a multi-stage object detection pipeline with Faster R-CNN & Multi-task loss, trained on **50K images**
- Implemented core detection components including RPNs, RoI pooling, and NMS for scalable multi-object inference
- Statistically concluded a **120% increase** in post-COVID pedestrian activity and delivered a deployable model

ML Engineer Intern

Aug 2022 – Jan 2023

Renzoe Box

Austin, TX

- Improved **skin-tone detection** algorithms, contributing to the Renzoe-Match engine
- Integrated **YOLOv4** and **dlib-based** facial detection pipelines for **120%** faster image processing
- Wrote web-scraping scripts with **BeautifulSoup**, improving product recommendation accuracy
- Constructed and preprocessed a **1000+ image** dataset for model training

PROJECTS

Autonomous Drone Delivery Pipeline | *Pytorch, Numpy, Matplot, ROS, A**

Jan 2025 – May 2025

- Built a CV landing decision pipeline using depth sensing, 3D point clouds, and RANSAC plane fitting
- Automated geometric reasoning over surface normals, slope, and flatness with **A*** and **RRT** path planning
- Collaborated with ROS and hardware teams to integrate perception with flight control, reducing landing-zone failures through iterative field testing

Credify.ai | *Fetch.ai uAgents, Validation Pipeline, GPT-4o, RAG, MV3, Claude*

Aug 2025 – Oct 2025

- Built a real-time credibility scoring system for Reddit posts using live context extraction
- Designed a **multi-agent backend** using Fetch.ai uAgents with gateway orchestration and NYT-based retrieval
- Implemented **LLM-based credibility scoring** with JSON schemas and exposed results via REST API
- Built an **MV3 extension** to inject a **React** dashboard using MutationObserver, tab listeners, and cached analysis

BoilerFixIt | *CI/CD, React, Node.js, Express, MongoDB, JWT, Stripe*

May 2025 – Sept 2025

- Co-built the Node.js/Express backend with **MongoDB, JWT auth, Stripe** & automated emails
- Designed **RESTful APIs** to manage user accounts, service requests & payments
- Designed & Tested **JSON schemas** to structure data models and API payloads
- Implemented a **React** frontend with authenticated routing

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, Bash, R, JavaScript, TypeScript, HTML/CSS

Frameworks: REST APIs, FastAPI, Flask, Node.js, Express, TensorFlow, PyTorch, React, WebSockets, ROS

Tools: Azure, Databricks, AWS (EC2, Lambda), Git, GitHub Actions, CI/CD, Docker, Kafka, Linux, GCP

Libraries: Pandas, NumPy, Matplotlib, LangChain, OpenCV, YOLO, Codex